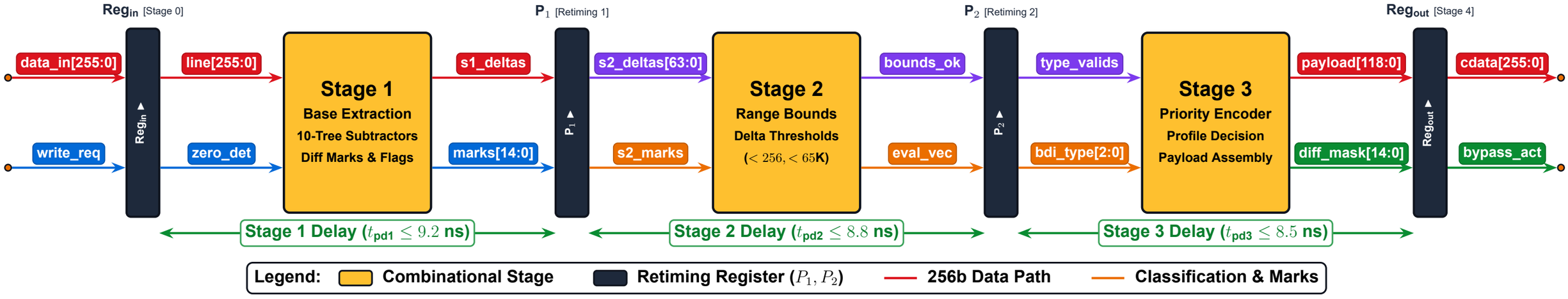
Reg_in [Stage 0]
P_1 [Retiming 1]
P_2 [Retiming 2]
Reg_out [Stage 4]
data_in[255:0]
write_req
Reg_in
line[255:0]
zero_det
Stage 1
Base Extraction
10-Tree Subtractors
Diff Marks & Flags
s1_deltas
marks[14:0]
P_1
s2_deltas[63:0]
s2_marks
Stage 2
Range Bounds
Delta Thresholds
(< 256, < 65K)
bounds_ok
eval_vec
P_2
type_valids
bdi_type[2:0]
Stage 3
Priority Encoder
Profile Decision
Payload Assembly
payload[118:0]
diff_mask[14:0]
Reg_out
cdata[255:0]
bypass_act
Stage 1 Delay (t_pd1 ≤ 9.2 ns)
Stage 2 Delay (t_pd2 ≤ 8.8 ns)
Stage 3 Delay (t_pd3 ≤ 8.5 ns)
Legend:
Combinational Stage
Retiming Register (P_1, P_2)
256b Data Path
Classification & Marks